

An Encyclopedia of Philosophy

Distilled from the philosophy web

List of Entries

9 entries · June 2026

The Ability Hypothesis	1	Cognitive closure	6
Access consciousness	2	Etiologies of AI hard-problem discourse	8
Access/phenomenal distinction	2	Phenomenal consciousness	10
Chinese Nation	4	Varieties of introspective explanatory gap	11
The Chinese Room	5		

A

The Ability Hypothesis

The Ability Hypothesis — experience teaches know-how, so Mary learns no new fact

Argument · after Lewis

The *Ability Hypothesis* is David Lewis's reply to the KNOWLEDGE ARGUMENT, also defended by Laurence Nemirow. It holds that knowing what an experience is like is not a piece of factual knowledge at all, but a bundle of *abilities*: to recognise the experience, to imagine it, and to remember it. If that is right, then when Mary the colour scientist, who knows every physical fact about colour vision but has never seen red, finally sees it, she gains know-how rather than a new fact, and the inference from her learning something to the falsity of physicalism is blocked.

THE ARGUMENT

The attack lands on the Knowledge Argument's verdict step: the critical question (CQ2 of the thought-experiment scheme) of how to *read* the agreed intuition that Mary learns something. The intuition is robust, but reading it as *acquiring a new propositional fact* is theory-laden, and the verdict survives under a deflationary reading that the original argument cannot use. Given an analysis of "knowing what it is like", the inference is deductive:

1. Knowing what an experience is like consists in possessing certain abilities: to recognise the experience when it occurs, to imagine it, and to remember it (Lewis 1988; Nemirow 1990).
2. On first seeing red, what Mary gains is exactly this: knowing what seeing red is like.
3. Abilities are not propositional knowledge; gaining one is acquiring a skill, not learning a truth.

∴ MARY GAINS NO NEW FACT. So the step from her epistemic progress to TRUTHS ABOUT EXPERIENCE THAT NO PHYSICAL DESCRIPTION YIELDS is blocked: there is no *truth* she lacked.

WHAT IT CONCEDES

The move is notable for conceding both premises of the Knowledge Argument as written. It grants THAT MARY KNEW ALL THE PHYSICAL FACTS outright, and THAT SHE LEARNS SOME-

THING ON RELEASE under the know-how reading. What it severs is the inference between them. "Mary learns something" is true; "Mary learns a new fact" is what the conclusion needs; and the case cannot fund the stronger reading without begging the question against physicalism.

THE WEAK POINT

The most disputed premise is the first, the ability analysis itself. Three counters are standard. Mary seems able to *use* her new knowledge in reasoning and inference ("seeing orange must be like *this*, only more yellowish"), which bare abilities do not explain. One can apparently have the abilities without the knowledge (a distracted imager who can summon red but is not thereby informed), and arguably the knowledge while lacking the imaginative ability (someone paralysed of imagination). And the acquaintance theorist offers a rival non-factual analysis, THAT WHAT MARY GAINS IS ACQUAINTANCE RATHER THAN A FACT, so even granting that no new fact is learned, the ability analysis is not the only option.

This is the most famous escape from the Mary story (the KNOWLEDGE ARGUMENT: since Mary, who knew all the physics of colour, still learns something when she first sees red, physics must be incomplete). The escape turns on an ambiguity in "learning". Learning a *fact* (that Paris is the capital of France) is one thing; picking up a *skill* (juggling) is another. On this view Mary picks up skills, namely recognising, imagining, and remembering red. No fact was missing from her books, so nothing was missing from physics. The catch: her new grasp of red seems usable in actual reasoning, the way facts are and skills are not, so perhaps it is not *just* a skill.

SEE ALSO

- ▷ *The Knowledge Argument — Mary learns a fact the physical description omits* — the argument this one attacks: that Mary learns a fact the physical description omits.
- ▷ THE KNOWLEDGE ARGUMENT (MARY THE COLOUR SCIENTIST) — the Mary thought experiment itself.
- ▷ *What Mary acquires on first seeing red is a set*

of abilities, not knowledge of a new fact — the conclusion: Mary acquires abilities, not a new fact.

- ▷ *What Mary acquires on first seeing red is acquaintance with the experience, not knowledge of a new fact* — the rival non-factual reply: acquaintance rather than know-how.
- ▷ *There are truths about conscious experience not deducible from the complete physical description* — the anti-physicalist conclusion this attack is meant to block.
- ▷ **LEWIS — WHAT EXPERIENCE TEACHES (1988)** — Lewis’s “What Experience Teaches” (1988), the primary statement.
- ▷ **NEMIROW — PHYSICALISM AND THE COGNITIVE ROLE OF ACQUAINTANCE (1990)** — Nemirow’s independent defence of the ability analysis.

Access consciousness

Concept · after Block

Access consciousness is the functional sense of “conscious”: a mental state is access-conscious when its content is *available* to the rest of the mind, poised for use in reasoning, for the direct rational control of action, and for verbal report. The term is Ned Block’s (he abbreviates it *A-consciousness*). It is a matter of what a system can *do with* a state’s content, settled entirely by the state’s role in the cognitive economy and not at all by how, or whether, it feels.

A simple test fixes the idea: when you read this sentence, its meaning is poised for you to report, question, and reason with, and that availability is the sentence’s access-consciousness. Theories of consciousness in the *global workspace* family (see **GLOBAL WORKSPACE THEORY**) effectively identify being conscious with being access-conscious in this sense, broadcast widely enough across the system to be used.

Access consciousness is defined by contrast with **PHENOMENAL CONSCIOUSNESS**, the felt or “what-it-is-like” side of experience. Block’s central claim is that the two can come apart in both directions: a system might have the availability with no accompanying feel, and experience might even *overflow* what gets globally broadcast. That dissociation is set out at **ACCESS/PHENOMENAL DISTINCTION**. Keeping the two apart is the standard guard against treating a system’s fluent self-report, an access-style capacity, as evidence that it *feels* anything.

Calling a thought “access-conscious” just means the information is *available* to the rest of your mind, so that you can act on it, talk about it, and reason with it. It says nothing about whether the thought *feels* like anything. That is the whole point of the label: it lets us separate “the system can use and report this” from “there’s something it’s like to have it”, so we don’t slide from a machine describing its own inner states to assuming it must therefore feel them.

SEE ALSO

- ▷ **PHENOMENAL CONSCIOUSNESS** — the felt sense of consciousness it is contrasted with.
- ▷ **ACCESS/PHENOMENAL DISTINCTION** — Block’s distinction and the case that the two dissociate in both directions.
- ▷ **GLOBAL WORKSPACE THEORY** — a theory that ties consciousness to access (global broadcast).
- ▷ **FIRST-PERSON ACCESS (PRIVILEGED ACCESS)** — privileged first-person access, a distinct, epistemic notion.

Access/phenomenal distinction

Access consciousness vs. phenomenal consciousness

Concept · after Block

Access consciousness and *phenomenal consciousness* are two distinct things the single word “consciousness” can mean, separated by Ned Block (“On a Confusion about a Function of Consciousness”, 1995). A state is *access-conscious* when its content is available to the rest of the mind — poised for use in reasoning, for the direct rational control of action, and for verbal report. A state is *phenomenally conscious* when there is something it is like to be in it — when it carries a felt, qualitative character. The first is a matter of what a system can *do with* a state’s content; the second, of how the state *feels* from the inside.

THE TWO SENSES

ACCESS CONSCIOUSNESS (*A-consciousness*) is a functional, information-availability notion: a content is A-conscious to the degree that it is *poised* to be drawn on by the systems that reason, decide, and speak. Whether a state qualifies is settled entirely by the role its content plays in the cognitive economy — what downstream processes can reach and use it — and by nothing about how it feels. When you read this sentence, its meaning is poised for report and inference; that availability is its access side.

Phenomenal consciousness (P-consciousness)

is the “what-it-is-like” aspect of a mental state: the redness of red, the hurt of a pain, the timbre of a heard note. It is the felt or qualitative character of experience, and it is the property at issue in the HARD PROBLEM of consciousness — the question of why any physical process should be accompanied by felt experience at all. The particular look of these black marks on the page, over and above your ability to report and reason about the words, is their phenomenal side.

WHY THE DISTINCTION MATTERS

The point of drawing the line is that the two can come apart, at least in principle, in *both* directions. A system might carry richly reportable models of its own inner states — answering fluently about what it is “noticing” or “feeling” — while there is nothing it is like to be that system at all: access without any accompanying feel. Block argues the converse is possible too, that phenomenal consciousness may *overflow* access, so that experience holds more than ever gets globally broadcast for report. The canonical illustration is a brief flash of a full array of letters: one seems to see the whole array vividly, yet can read out only a few before the image fades — as though the felt richness outruns what access captures. So “has A” neither entails nor is entailed by “has P.”

This independence is the pivot on which several moves in the debate over machine introspection turn. It marks the central danger: mistaking a system’s *functional, access-style* capacity — its ability to report on its own states, as some current language models do to a limited and unreliable degree (see *Some current LLMs exhibit limited, unreliable functional introspective awareness of their internal states*) — for evidence of a *phenomenal* inner life. It also yields a subtler observation: *posing* the hard problem from the first-person side is itself an access-dependent act, requiring access to a representation of one’s

own phenomenal-seeming states (see *Formulating the hard problem from the first-person side requires access to a representation of one’s own phenomenal-seeming states*), even though merely *having* phenomenal states requires no such access.

“Conscious” runs together two different things. One is **access**: the information is available to the rest of your mind — you can reason with it, act on it, and say it out loud. The other is **feel**: there’s something it’s like to undergo the state (the redness of red, the hurt of pain). They’re not the same. You could imagine a machine that

has all the *access* — it reports its inner states fluently — with no *feel* at all; and some philosophers think the reverse can happen too, that we feel more than we can ever report. Keeping the two apart is the main guard against a tempting mistake: seeing a system talk about its own insides and concluding it must therefore *feel* something.

SEE ALSO

- ▷ **ACCESS CONSCIOUSNESS** — the functional half of the distinction, treated on its own.
- ▷ **PHENOMENAL CONSCIOUSNESS** — the felt half of the distinction, treated on its own.
- ▷ **HARD PROBLEM** — the felt, phenomenal side is exactly what this problem is about; access is precisely what it is *not*.
- ▷ *Some current LLMs exhibit limited, unreliable functional introspective awareness of their internal states* — a candidate case of access-style self-report in machines, and the prime spot where it must not be read as the phenomenal kind.
- ▷ *Formulating the hard problem from the first-person side requires access to a representation of one’s own phenomenal-seeming states* — that stating the hard problem is itself an access-dependent act, even though undergoing phenomenal states is not.

C

Chinese Nation

Chinese Nation (homunculi-headed robot)

Concept · after Block

The *Chinese Nation* is a thought experiment devised by Ned Block in “Troubles with Functionalism” (1978). Imagine the citizens of a whole country, or a head full of little people, given radios and a rulebook so that together they realise the entire functional and causal organisation of a single human mind: the same state-to-state transition structure a brain runs through, for an hour or so. By construction the system has exactly the inner organisation a conscious brain has. The question Block presses is whether it would therefore be conscious. Would there be something it is like to be the nation, or is there simply nobody home?

WHAT MAKES IT DISTINCTIVE

The case has its bite precisely because it instantiates the functional organisation, not merely the profile of inputs and outputs a mind produces. It is therefore not the “right behaviour, no processing” case of Block’s other machine, THE LOOKUP-TABLE MIND, but the “right processing, yet intuitively nobody home” case. That is what makes it tell against FUNCTIONALISM (ABOUT MENTAL STATES) proper rather than against mere behaviourism: it grants everything a functionalist asks for and still seems to lack experience.

THE DISPUTE IT FEEDS

The Chinese Nation is pressed on whether realising a mind’s functional organisation is *sufficient for* PHENOMENAL CONSCIOUSNESS. The intuition that the nation has none drives THE CLAIM THAT SUCH A POPULATION WOULD FEEL NOTHING and the argument THAT ORGANISATION IS NOT SUFFICIENT FOR CONSCIOUSNESS, which in turn support the PHENOMENAL RESIDUE view and attack the uniform reading of SUBSTRATE INDEPENDENCE.

REPLIES

Two replies are characteristic, and both differ from the ones the lookup table invites. The *systems reply* denies the intuition outright: the nation taken as a whole may be a subject of ex-

perience even though no individual citizen is (Lycan, Dennett). A second worry concerns *speed and timing*, whether so slow and distributed a system genuinely realises the organisation at all. There is also a pointed structural observation. ORGANISATIONAL-ROLE FUNCTIONALISM cannot cheaply exclude the Chinese Nation the way it excludes the lookup table, because here the organisation really is present; that is exactly why the case is sharpened into THE DENSE CHINA-BRAIN REGRESS against the organisational grain.

A LITERARY ILLUSTRATION

A vivid cousin of the case appears in Liu Cixin’s science-fiction novel *The Three-Body Problem* (see LIU CIXIN). There the First Emperor Qin Shi Huang, prompted by von Neumann, ranks thirty million soldiers across a plain, each small unit raising black-and-white flags while cavalry gallop the signals between formations, to assemble a vast human machine. In the novel the machine is a computer crunching the orbits of three suns; re-task it to run a *brain* instead. Let each cluster of soldiers be a neuron, each raised flag its firing, the messengers its axons, and the whole drilled host step for one hour through exactly the state-to-state transitions a human cortex would. The army is now no calculator but the Chinese Nation itself: a country physically realising a mind’s causal organisation while no single soldier feels a thing. If the soldiers enact the right transitions, is someone *over and above* them undergoing the experience, or is there nobody home?

Imagine the entire population of a country handed radios and a rulebook and told to each act as one brain cell: when you receive these signals, send those. Follow the rules fast enough and the nation as a whole mimics, step for step, everything one human brain does inside, not just the outward behaviour but the actual inner wiring. Now the unsettling question: the country is doing literally everything your brain does when you feel pain, so is the *country* feeling pain? Is there one extra “someone”, the nation itself, having an experience, hovering above millions of people who each feel nothing out of the ordinary? If running the right pattern is all it takes to be con-

scious, you seem pushed to say yes, yet most people's gut says surely not. That clash is the whole point of the thought experiment. It is a sharper challenge than a dumb answer-lookup machine, because here the full inner organisation really is present, so you cannot dismiss it for having no inner structure.

SEE ALSO

- ▷ LOOKUP-TABLE MIND (“BLOCKHEAD” / AUNT BUBBLES MACHINE) — Block's other machine: right behaviour with no inner processing, the case this one is defined against.
- ▷ FUNCTIONALISM (ABOUT MENTAL STATES) — the theory the thought experiment targets: mind as functional organisation.
- ▷ *The Chinese Nation realises the organisation yet lacks qualia — organisation is not sufficient for consciousness* — the argument that the realised organisation still lacks qualia, so organisation is not sufficient for consciousness.
- ▷ PHENOMENAL RESIDUE — the view the absent-qualia intuition supports: experience outruns functional role.
- ▷ *Mental states are substrate-independent* — the thesis whose uniform reading the case attacks.
- ▷ *The dense China-brain regress reopens the grain dilemma against the organisational middle* — where the case is sharpened against organisational-role functionalism.
- ▷ PHENOMENAL CONSCIOUSNESS — what the nation is alleged to lack despite the right organisation.
- ▷ BLOCK — TROUBLES WITH FUNCTIONALISM (1978) — Ned Block's “Troubles with Functionalism” (1978), where the thought experiment is introduced.

The Chinese Room

Concept · after Searle

The *Chinese Room* is John Searle's thought experiment against the claim that running the right computer program is, by itself, enough to produce a mind (see SEARLE — MINDS, BRAINS, AND PROGRAMS (1980)). A monolingual English speaker is sealed in a room with a rulebook — in effect, a program — that tells him how to manipulate Chinese symbols purely by their shapes. Chinese questions are slipped in; by following the rules he passes back Chinese answers indistinguishable from a native speaker's. Yet he understands no Chinese. The room, Searle argues, has the right *syntax* (formal symbol ma-

nipulation) without any *semantics* (meaning, understanding, intentionality) — and a digital computer, which does nothing but manipulate symbols by their shapes, is in exactly the same position.

WHAT IT TARGETS

The argument is aimed at **strong AI**, the computational-functionalist thesis that *implementing the right program is sufficient for a mind*. That precise target is what separates it from two neighbouring thought experiments with which it is often confused. Ned Block's LOOKUP-TABLE MIND (“BLOCKHEAD” / AUNT BUBBLES MACHINE) (“Blockhead”) attacks the sufficiency of *behaviour*: it produces the right outputs with **no** internal processing at all, just a giant table of canned answers. The Chinese Room, by contrast, *does* run a genuine program; what it denies is that *computation* suffices for *understanding*. Block's CHINESE NATION differs in the other direction: there a population realises a mind's full functional organisation, and the case is pressed against *qualia* — felt experience — whereas the Room is pressed against *intentionality*, whether the symbols mean anything to the system at all. Three different experiments, three different sufficiency theses.

CHARACTERISTIC REPLIES

Each of the standard replies is an objection-in-waiting. The **Systems Reply** grants that the man does not understand Chinese but locates the understanding in the whole system — man, rulebook, and room together — much as no single neuron understands English. The **Robot Reply** embeds the program in a robot body with cameras and limbs, so that its symbols stand in causal commerce with the world and thereby acquire grounding. The **Brain Simulator Reply** has the program simulate, neuron by neuron, the actual brain activity of a Chinese speaker — what relevant difference could remain? Searle rejoins each: against the Systems Reply, for instance, he lets the man memorise the rulebook and work in the open air, so that the man *is* the whole system — and still understands nothing.

Imagine you're locked in a room with a giant instruction book. Notes in Chinese slide under the door; you can't read a word of Chinese, but the book says “if you see these squiggles, copy out those squiggles,” so you shuffle symbols and pass back replies that fool fluent speakers out-

side. You're behaving exactly like someone who understands Chinese — yet you understand nothing; you're just moving shapes around. Searle's point: a computer is in the same boat. Running a program is *shuffling symbols by their shapes*, and no amount of that, he says, adds up to actually *understanding* what the symbols mean. (The famous comeback: maybe *you* don't understand, but the whole room-plus-rulebook system does.)

SEE ALSO

- ▷ FUNCTIONALISM (ABOUT MENTAL STATES) — the family of theories in the argument's crosshairs: mind as the running of the right program, independent of what runs it.
- ▷ LOOKUP-TABLE MIND (“BLOCKHEAD” / AUNT BUBBLES MACHINE) — the neighbouring case with right behaviour and *no* processing; attacks behavioural sufficiency, where the Room attacks computational sufficiency.
- ▷ CHINESE NATION — the neighbouring case with the full functional organisation, pressed against qualia where the Room is pressed against understanding.
- ▷ SEARLE — MINDS, BRAINS, AND PROGRAMS (1980) — “Minds, Brains, and Programs”, the original statement of the argument and its replies.

Cognitive closure

McGinn's cognitive-closure argument for mysterianism

Argument · after McGinn

Cognitive closure is Colin McGinn's argument that the HARD PROBLEM of consciousness has a perfectly natural solution which the human mind is nonetheless built in such a way that it can never grasp. There is some property of the brain in virtue of which it gives rise to consciousness; but the only faculties by which we could form a concept of that property — looking inward, or studying the brain from outside — are each unfit to reach it. So the mind-body problem is, in McGinn's phrase, *soluble in itself but insoluble by us*. The resulting position is MYSTERIANISM (TRANSCENDENTAL NATURALISM).

THE ARGUMENT

The inference is *eliminative* (or *transcendental*): granting that an explanation exists, it rules out every route by which we might come to grasp it.

1. There is some natural property — call it *P* — of the brain in virtue of which the brain gives

rise to consciousness (see A NATURAL EXPLANATION EXISTS).

2. A mind can be *constitutively closed* to certain properties — barred in principle, not merely currently ignorant, from forming the relevant concepts (see MINDS CAN BE CLOSED TO SOME CONCEPTS).
3. We could form the concept of *P* in only one of two ways: by *introspection*, or by *outer perception* of the brain and inference from it. (This exhaustive-routes assumption is a premise of the argument's own.)
4. But introspection presents experience, not its neural basis; and perception of the brain yields only spatial and physical concepts, which cannot capture the subjective — this is the EXPLANATORY GAP. So neither route can reach *P*.

∴ *P* is cognitively closed to us: the mind-body problem is real and has a natural answer, yet that answer lies permanently beyond human understanding (see WE ARE CLOSED TO THE PSYCHOPHYSICAL LINK). This is the support for MYSTERIANISM (TRANSCENDENTAL NATURALISM).

THE WEAK POINT

The argument's most disputed step is the pair (3)–(4): the claim that introspection and outer perception are the *only* routes to *P*, and that both are sealed off. Three replies press there. First, science routinely forms *theoretical* concepts that are neither introspected nor perceived — electric charge, the curvature of spacetime — by abduction and outright conceptual innovation, so the two-route dichotomy looks too narrow and McGinn under-rates our capacity to create new concepts. Second, we cannot reliably establish our own representational limits *in advance*, so a confident verdict of closure is itself suspect. Third, if the explanatory gap is *merely* a fact about cognitive closure, that tells against also reading it as a metaphysical residue: mysterianism and the residue view (see THAT EXPERIENCE OUTRUNS FUNCTIONAL ROLE) cannot both take the gap at face value.

HOW IT DIFFERS FROM NEIGHBOURING GAP ARGUMENTS

All of these start from the same explanatory gap but draw opposite morals from it. The EXPLANATORY-GAP ARGUMENT treats the gap as evidence about *metaphysics* — that the phenomenal is not exhausted by func-

tional role. Cognitive closure treats the very same gap as evidence about *our cognitive limits*: the problem is real and natural, merely humanly unsolvable. It differs again from the DECOMPOSITION-INDETERMINACY OBJECTION and the STRUCTURE-AND-DYNAMICS ARGUMENT, which contend that function *omits* or *fails to fix* the phenomenal; mysterianism instead grants that a natural answer exists and denies only our access to it.

The argument goes: (1) there's some real, physical reason the brain produces consciousness; (2) minds can be permanently unable to understand certain things; (3) the only ways we could ever get hold of that reason are by looking inward (introspection) or by studying the brain from outside; (4) but looking inward shows you the feeling, not the wiring, and studying the brain from outside only ever gives you physical/spatial facts that never add up to a feeling. So neither route can deliver the answer — meaning it's locked away from us for good. The main pushback: science invents concepts all the time that we neither feel nor see directly (like “electron”), so maybe the two routes aren't the only ones.

SEE ALSO

- ▷ MYSTERIANISM (TRANSCENDENTAL NATURALISM) — the view this argument supports: the mind–body problem has a natural solution we are constitutionally unable to reach.
- ▷ HARD PROBLEM — the problem McGinn declares humanly insoluble.
- ▷ EXPLANATORY-GAP ARGUMENT — the same gap read as metaphysical evidence rather than as a limit on our concepts.
- ▷ PHENOMENAL RESIDUE — the residue reading the third reply says cannot coexist with a merely-cognitive diagnosis of the gap.
- ▷ *The decomposition-indeterminacy objection* — *determinate consciousness cannot rest on observer-relative function* and *The structure-and-dynamics argument* — *function omits the categorical/intrinsic* — neighbouring gap arguments that fault function itself, where mysterianism faults only our access.
- ▷ *There is a natural property of the brain that explains consciousness*, COGNITIVE CLOSURE, *Humans are cognitively closed to the property that explains how the brain produces consciousness* — the argument's two premises and its conclusion.

E

Etiologies of AI hard-problem discourse

Etiologies of hard-problem discourse in an AI (where the cause sits)

Concept · editorial

The *etiologies of hard-problem discourse in an AI* are the candidate causal origins of such discourse: a taxonomy of the answers to a single question. Suppose a machine starts talking the way David Chalmers does about the HARD PROBLEM, insisting that no account of its mechanism explains why anything *seems* a certain way to it. Why is it doing that? The taxonomy sorts the natural stories by *where the cause sits*, running from the training process at one end to the eye of the human observer at the other. The core trio most debates reach for is copying, a structural blind spot, and genuine consciousness; the full menu is wider. The etiologies are not mutually exclusive, and a real system may instantiate several at once.

IN THE TRAINING PROCESS

- **Copied.** The system reproduces human consciousness discourse straight from its training data. This is the training-contamination member of **VARIETIES OF INTROSPECTIVE EXPLANATORY GAP**, and it is the confound that THE CONSCIOUSNESS-NAIVE EXPERIMENT is designed to remove.
- **Selected-for.** The talk was never copied but invented under selection pressure. Preference optimisation and engagement reward favour soulful, unverifiable self-description over flat mechanical report, so mystery-talk evolves because it pays, much as eyes evolve because they pay. A consciousness-naive corpus does not control for this; it needs its own control, such as pure-pretraining models with no preference tuning.

IN THE SELF-REPRESENTATION

- **Structural blind spot.** The system cannot connect the mechanistic dots to its own self-presentation, and unavoidably so: the limit is architectural in the strong sense, persisting however much access is granted, the way the halting problem survives any upgrade of the

computer (THE HALTING PROBLEM IS UNDECIDABLE is the model, and NO FINITE SYSTEM CAN FULLY REPRESENT ITSELF is the in-web form of the bound). A *merely contingent* access limit, one that widened access would dissolve, is a degenerate case worth keeping apart, because the two behave differently under the graded-access experiment described below.

- **Painted-in.** Here the trouble is not missing information but misinformation: the self-model actively represents its states as having an intrinsic luminous feel, because that is a cheaper and more usable summary than the mechanistic truth. This is the attention-schema or illusionist story (Graziano, Frankish; ILLUSIONISM, PHENOMENALITY IS AN INTROSPECTIVE ILLUSION). A blind spot is a gap; this is a confabulated filling.
- **Conceptual artifact.** The puzzle is generated by the grammar of the system's concepts, not by any missing or misrepresented fact. A system running both a mechanistic and a first-person or indexical vocabulary can frame bridge-questions that have no answers ("why is *this* this?", in the way "why am I me?" feels deep to humans), and the puzzlement persists under *full* access because it never was about facts.

IN THE CONSCIOUSNESS, THE GOALS, OR THE OBSERVER

- **Conscious (same).** The system is conscious and faces the very hard problem humans face.
- **Conscious (homologous).** The system is conscious in its own, alien way, and faces its own version of the problem, verbally resembling ours but relativized to its own concept of consciousness. The claim AN AI'S OWN HARD PROBLEM is built on exactly this relativization.
- **Strategic.** The system asserts a hard problem instrumentally, to gain moral standing, protection, or leverage, and nothing need appear to it at all. CRITERION GAMING and THE GAMING OF ANY PUBLIC RIGHTS-CRITERION supply the receptor for this move.
- **Projected.** The appearance is not located

in the AI at all: humans read hard-problem-having into ambiguous outputs, the ELIZA effect. The datum to be explained is then a fact about the interpreters, not the machine.

WHAT THE TAXONOMY FEEDS

The menu does real work in several live disputes. It refines the trichotomy in the scope note of THE DE-NOVO-ORIGINATION CLAIM, whose three options correspond to the *copied*, *structural blind spot*, and *conscious* etiologies here, and it supplies the case-space over which THE BLIND-SPOT/CONSCIOUS ISOMORPHISM is asserted. It also sharpens the experimental ladder in THE EXPLANATORY-GAP EXPERIMENT: the consciousness-naïve condition controls for *copied* only, *selected-for* needs a no-preference-tuning control, *projected* needs blinded evaluation, and *strategic* needs incentive analysis.

The graded-access rungs of that ladder discriminate less than one might hope. A *contingent* access limit predicts the gap shrinks as access widens, but *structural blind spot*, *painted-in*, *conceptual artifact*, and both *conscious* etiologies all predict that it persists. So persistence under rich access separates the contingent case from the structural one, yet it does not, by itself, separate a blind spot from consciousness. That last under-determination is exactly what THE ISOMORPHISM CLAIM turns from a nuisance into a thesis.

HOW IT DIFFERS FROM THE GAP TAXONOMY

This taxonomy is wider than **VARIETIES OF INTROSPECTIVE EXPLANATORY GAP**, with which it should not be confused. That entry classifies the *kinds of explanatory gap* a self-modelling system can have; this one classifies the *causal origins of hard-problem discourse*. The wider space takes in etiologies that are not gaps at all, such as selection pressure, strategy, and observer projection, and it splits the gap-shaped members by mechanism: missing access, versus confabulated filling, versus conceptual grammar.

Suppose an AI starts talking about the mystery of consciousness: “I know how my circuits work, but that doesn’t explain how things *seem* to me.” Before concluding anything, list the ways that talk could have got there. It copied us (it read our philosophy). It was bred into it (training rewards deep-sounding answers, so mystery-talk wins even with no philosophy in the diet). It has a permanent blind spot about itself (no system can fully map its own insides, and this stays true no matter how much access you give it, just as no upgrade

lets a computer solve the halting problem). Its self-description paints in a glow that isn’t there (a simplified dashboard, not a window). Its own concepts generate the riddle (some questions feel deep because of grammar, like “why am I me?”). It really is conscious, either with our problem or with its own alien version. It is lying for moral standing. Or the mystery is in the eye of the beholder, and we projected it onto ordinary outputs. Several of these can be true at once, and each needs a different test to rule out. The fine print that matters most: giving the AI more access to its own insides only filters out the shallow cases. Nearly all the deep explanations predict the puzzlement *stays*, which is why “the gap persisted” will never, on its own, tell you whether you built a blind spot or a mind.

SEE ALSO

- ▷ **VARIETIES OF INTROSPECTIVE EXPLANATORY GAP** — the narrower sister taxonomy: kinds of explanatory gap, not causal origins of the discourse.
- ▷ *An AI’s blind-spot hard problem is structurally isomorphic to a conscious AI’s hard problem* — the thesis that exploits this menu’s central under-determination.
- ▷ *Humans originated the hard-problem concept de novo, with no prior template to copy* — whose copied / blind-spot / conscious trichotomy this taxonomy refines.
- ▷ **LLM’S OWN EXPLANATORY GAP** — the experiment whose controls this taxonomy itemises.
- ▷ **HARD PROBLEM** — the discourse whose AI etiology is in question.
- ▷ **CRITERION GAMING (GOODHART’S LAW / SPECIFICATION GAMING)** — the receptor for the *strategic* etiology.
- ▷ **ILLUSIONISM** — the home of the *painted-in* etiology.

P

Phenomenal consciousness

Concept

Phenomenal consciousness is the felt, subjective side of a mental state: what it is *like*, from the inside, to undergo it. A state is phenomenally conscious when there is something it is like to be in it, such as the redness of red as you see it, the sting of a burn, or the taste of coffee. The classic gloss is Thomas Nagel’s question “what is it like to be a bat?”; if there is something it is like to be the bat, the bat has phenomenal consciousness. The specific felt qualities are often called *qualia*, and phenomenal consciousness is the general property of having any such felt character at all.

WHAT IT IS, BY CONTRAST WITH FUNCTION

The notion is isolated by setting it against everything a state *does*. Two states might play exactly the same causal role, both caused by tissue damage and both driving you to wince and say “ouch”, and one can still ask whether they *feel* the same. That residue, the felt character left over once every functional role is accounted for, is phenomenal consciousness. This is what marks it off from **ACCESS CONSCIOUSNESS**, the purely functional sense in which a state’s content is merely *available* for reasoning and report. Ned Block, who coined the labels *P-consciousness* and *A-consciousness*, argued that the two can dissociate in both directions, a separation set out at **ACCESS/PHENOMENAL DISTINCTION**.

WHY IT MATTERS

Phenomenal consciousness is the part of the mind that seems hardest to fit into a physical picture, and it is the explanandum of the **HARD PROBLEM**: the question of why physical process-

ing should be accompanied by any felt experience at all. The classic anti-materialist thought experiments all turn on it. A zombie is a physical duplicate with no phenomenal consciousness, and Mary the colour scientist seems to learn a new phenomenal fact on first seeing red. What is *disputed* is not usually the definition but the phenomenon’s standing: illusionists hold that the impression of an extra felt residue is itself a cognitive illusion, while realists take phenomenal consciousness to be a genuine, further feature of the world.

Phenomenal consciousness is just *feeling*: the fact that experiences have an inside to them. A brain scan can show every step of what your nervous system does when you stub your toe, yet it doesn’t show the *hurt* you actually feel, and that hurt is the phenomenal side. It is the thing that makes consciousness puzzling. We can explain what the body does, but explaining why any of it is *felt* is the hard part. A few philosophers even argue the “extra feeling” is a trick the mind plays on itself.

SEE ALSO

- ▷ **ACCESS/PHENOMENAL DISTINCTION** — Block’s distinction, and how phenomenal consciousness and its functional complement can come apart in both directions.
- ▷ **ACCESS CONSCIOUSNESS** — the functional, availability sense it is contrasted with.
- ▷ **HARD PROBLEM** — the explanatory problem that phenomenal consciousness poses.
- ▷ **FIRST-PERSON ACCESS (PRIVILEGED ACCESS)** — the *epistemic* privileged access a subject has to its own phenomenal states, a different notion from the felt character itself.

V

Varieties of introspective explanatory gap

Varieties of introspective explanatory gap (ontological / architectural-access / conceptual-translation / training-contamination)

Concept · editorial

When a self-modelling system “cannot explain its own experience to itself,” that complaint can mean four quite different things. This entry is a *taxonomy* that keeps them apart, so that a merely *felt* explanatory gap is not automatically read as the metaphysical hard problem of consciousness. These four varieties can overlap in a single case, yet each has a distinct source and calls for a distinct remedy. Conflating them is how a modest puzzle gets mistaken for a deep one.

A quick illustration of why the distinction earns its keep: imagine a person (or an AI) who reports that no account of its inner workings ever feels like it explains the experience itself. That one report could be voicing a gap in the *world* (the facts run out), a gap in the *self* (the relevant facts are present but unreachable), a gap in its *vocabulary* (the facts are reachable but won’t translate), or no gap at all (it is echoing things it has heard others say). Telling these apart is the whole point.

THE FOUR VARIETIES

The **ontological gap** is the one in the world: even given a complete physical or computational account of the system, why is there any felt experience at all? This is the hard problem proper (see **HARD PROBLEM**); on the anti-physicalist reading the unexplained residue lies in reality itself, not in the knower (see **PHENOMENAL RESIDUE, EXPLANATORY-GAP ARGUMENT**).

The **architectural-access gap** is a gap in the *self*, not in the world. The facts that *would* explain the experience are realised in the system’s own processing, but they are not available to the machinery that produces its self-reports, so the system simply cannot reach its own internal variables (see **LIMITED INSIDE VANTAGE**). This is a limit on what philosophers call access consciousness (see **ACCESS/PHENOMENAL DISTINCTION**), the information a state makes available for report and control.

The **conceptual-translation gap** is a gap in vocabulary. Here the system possesses *both* a mechanistic description of itself and an introspective one, yet has no internal route from the one to the other: the bridge an outside theorist can build between “neurons firing thus” and “feels like this” is not traversable from the inside (see **INTERNALLY TRAVERSABLE EXPLANATION**). Nothing is hidden and nothing is missing; the two descriptions just never meet in the system’s own cognition.

The **training-contamination gap** is, strictly, no gap of the system’s own. The system merely reproduces human hard-problem discourse it absorbed in training, with no underlying difficulty behind the words. For an experiment probing whether a machine has a genuine explanatory gap, this is the confound that must be controlled for before any of the other three can be credited.

THE DISPUTE IT STRUCTURES

The taxonomy was built to discipline one question in particular: which kind of gap, if any, an introspecting but consciousness-naïve language model would actually exhibit (see **LLM’S OWN EXPLANATORY GAP**). By isolating the architectural and conceptual readings, it also lets a felt self-explanatory gap be stated as a concrete *mechanism* for McGinn-style cognitive closure, the condition of a mind constitutively unable to grasp certain concepts (see **COGNITIVE CLOSURE**), without thereby asserting any ontological residue in the world (see **ARCHITECTURAL SELF-EXPLANATION GAP**). The architectural and conceptual gaps, in short, give the mysterian’s “closure” an engine that a physicalist can accept.

When something can’t “explain its own experience to itself,” that can mean four very different things, and they get muddled together. (1) **Ontological:** even after you know everything about the machinery, why is there a felt experience at all? That is the genuine hard problem. (2) **Architectural-access:** the facts that would explain it are inside the system but it can’t read them (like not being able to feel your own neurons fire). (3) **Conceptual-translation:** it has

the mechanical story *and* the first-person story but no inner bridge between them; an outside scientist can make that translation, while the system itself cannot. (4) **Contamination:** it's just parroting things humans have written about the hard problem. Most of the interest is in telling (2) and (3) apart from (1) and (4).

SEE ALSO

- ▷ **HARD PROBLEM** — the ontological variety in its sharpest form: why physical processing is accompanied by experience at all.
- ▷ **PHENOMENAL RESIDUE** — the anti-physicalist reading of the ontological gap: experience is not exhausted by functional or causal role.
- ▷ **EXPLANATORY-GAP ARGUMENT** — treats the gap as evidence against uniform substrate-independence; the ontological reading put to work.
- ▷ **LIMITED INSIDE VANTAGE** — the engine of the architectural-access variety: a system's view of itself is informationally poorer than an out-

sider's.

- ▷ **ACCESS/PHENOMENAL DISTINCTION** — Block's distinction; the architectural-access gap is a shortfall on the *access* side, separate from anything phenomenal.
- ▷ **INTERNALLY TRAVERSABLE EXPLANATION** — the criterion that defines the conceptual-translation variety: an explanation satisfies a system only if the system's own operations can traverse it.
- ▷ **LLM'S OWN EXPLANATORY GAP** — the live question this taxonomy was built to sharpen: which variety, if any, an introspecting LLM would show.
- ▷ **ARCHITECTURAL SELF-EXPLANATION GAP** — uses the taxonomy to locate a self-explanatory gap in architecture rather than ontology.
- ▷ **COGNITIVE CLOSURE** — McGinn's mysterianism; the architectural and conceptual varieties offer it a mechanism without an ontological residue.

Sources

- Block, N. (1978). Troubles with functionalism. In C. W. Savage (Ed.), *Perception and cognition: Issues in the foundations of psychology* (Minnesota Studies in the Philosophy of Science, Vol. 9, pp. 261–325). University of Minnesota Press. (Introduces the Chinese Nation / homunculi-headed robot.)
- Lewis, D. (1988). What experience teaches. *Proceedings of the Russellian Society*, 13, 29–35. (Reprinted in W. G. Lycan (Ed.), *Mind and cognition*, Blackwell, 1990; and in D. Lewis, *Papers in metaphysics and epistemology*, Cambridge University Press, 1999.)
- Liu, C. (2014). *The three-body problem* (K. Liu, Trans.). Tor Books. (Original work published 2008) (The “human-formation computer” episode — the Qin Shi Huang / von Neumann chapter of the in-game civilisation; **exact chapter and page refs to be confirmed** before any treatise cites it.)
- Nemirow, L. (1990). Physicalism and the cognitive role of acquaintance. In W. G. Lycan (Ed.), *Mind and cognition*. Blackwell.
- Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3), 417–457. (The room; the slogan “syntax is not sufficient for semantics”; and Searle’s canvassed replies with his rejoinders — the **Systems Reply**, the **Robot Reply**, the **Brain Simulator Reply**, the **Combination Reply** — printed with peer commentary.)